# Sentence Embeddings via Siamese Transformer Networks

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[My Code] [Template]

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## 1 Objectives and Dataset

Dense document and sentence vectorization constitutes a fundamental prerequisite in modern Information Retrieval, semantic deduplication, and vector database indexing (e.g., using libraries such as Faiss). The core objective of this assignment is to produce a standalone **Sentence Embedder**: a neural model that maps arbitrary, variable-length sentences into fixed-dimensional vectors $\mathbf{u} \in \mathbb{R}^d$ whose geometric proximity (cosine similarity) directly reflects semantic relatedness.

To achieve this, we investigate two distinct paradigms:

1. An unsupervised static **baseline** using token-level mean pooling over pre-trained GloVe vectors.
2. A simplified **Siamese Sentence-BERT (SBERT)** architecture [1]. Crucially, because sentence encoders cannot be trained without an objective signal, we employ a 3-class Natural Language Inference (NLI) classifier as an *auxiliary pretext task*. Training the network to classify whether sentence pairs are semantically aligned (*entailment*), neutral, or incompatible (*contradiction*) forces the underlying Transformer to sculpt a semantically meaningful embedding space. Once trained, the classification head is discarded, retaining solely the encoder for universal sentence vectorization.

### 1.1 The SNLI Corpus

We train and validate the embedder on the Stanford Natural Language Inference (SNLI) corpus [2], containing 570,152 sentence pairs. Each entry consists of an anchor premise ($s_1$) and an associated hypothesis ($s_2$) paired with one of three human-annotated ground-truth labels:

- **Entailment**: $s_2$ is necessarily true given $s_1$.
- **Neutral**: $s_2$ might be true given $s_1$, but lacks definitive entailment.
- **Contradiction**: $s_2$ is strictly false or logically mutually exclusive with $s_1$.

Instances lacking annotator consensus (gold label `'-'`) are pruned, yielding 549,367 valid training candidates. To ensure manageable training times on local compute, we utilize a representative 1% mini-corpus slice (`MINI_CORPUS = True`, `MINI_PERCENTAGE = 0.01`), comprising $N = 5{,}493$ pairs. The subset exhibits an empirical balance across the three target classes: 1,839 entailment, 1,821 neutral, and 1,833 contradiction instances.

## 2 Method and Program Structure

### 2.1 Summary of Reimers and Gurevych's SBERT

Standard cross-encoder BERT models concatenate pairs into a single sequence ([CLS] $s_1$ [SEP] $s_2$) and compute full cross-attention across all tokens. While highly accurate for pairwise classification, this approach suffers from $\mathcal{O}((n_1+n_2)^2)$ computational complexity, making similarity search over millions of candidate sentences computationally prohibitive (e.g., finding the closest match among 10,000 documents requires $\approx 50$ million forward passes).

To overcome this bottleneck, Reimers and Gurevych (2019) [1] introduced Sentence-BERT (SBERT). SBERT adopts a Siamese twin architecture where shared encoder weights project arbitrary-length sentences $s_1$ and $s_2$ independently into contextualized token states, which are subsequently aggregated via mean-pooling into fixed sentence vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$.

To learn meaningful representations without direct sentence-level supervision, SBERT employs a **3-way classification pretext training strategy**. The embeddings $\mathbf{u}$ and $\mathbf{v}$ are concatenated with their absolute element-wise difference:

$$\mathbf{h} = [\mathbf{u},\ \mathbf{v},\ |\mathbf{u} - \mathbf{v}|] \in \mathbb{R}^{3d} \tag{1}$$

where $|\mathbf{u} - \mathbf{v}|$ acts as an explicit distance metric, while $\mathbf{u}$ and $\mathbf{v}$ supply base semantic information. The vector $\mathbf{h}$ is passed into an auxiliary linear classifier:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}\mathbf{h} + \mathbf{b}), \quad \mathbf{W} \in \mathbb{R}^{3\times 3d} \tag{2}$$

Optimizing cross-entropy loss over this 3-class objective forces $\mathbf{u}$ and $\mathbf{v}$ to converge when sentences are logically equivalent and to diverge when contradictory. **Following training, the classification head W is discarded.** The standalone encoder is then used to map individual sentences into vectors, allowing semantic similarity to be computed instantaneously via cosine similarity:

$$\text{cosine}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \, \|\mathbf{v}\|_2} \tag{3}$$

## 2.2 Summary of the Baseline (GloVe Mean-Pooling)

As established in Section 4.1 and Table 1 of [1], a widespread unsupervised baseline computes sentence representations by taking the element-wise average of static word embeddings. We load the Stanford GloVe 6B dictionary [3] ($d = 50$, 400,000 words) augmented with reserved structural tokens: [PAD] = 0, [UNK] = 1, and [CLS] = 2.

For any sentence $s = (w_1, w_2, \ldots, w_T)$, the baseline sentence embedding $\mathbf{v}_{\text{base}}$ is computed as:

$$\mathbf{v}_{\text{base}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{E}(w_t) \tag{4}$$

where $\mathbf{E}(w_t)$ extracts the 50-dimensional GloVe embedding for token $w_t$. Semantic similarity across sentence pairs is computed directly via cosine similarity without any parameter updates.

## 2.3 Model Architecture and Structural Specification

Our PyTorch SBERT implementation replicates the architectural topology of [1] under an efficient configuration adapted to 50-dimensional word vectors. Table 1 details the exact layer-by-layer specifications of the model as inspected in the PyTorch module graph.

| Component | PyTorch Submodule Class | Dimensionality / Configuration | Parameters |
|---|---|---|---|
| **Embedding Layer** | `nn.Embedding` | 400,003 → 50 (GloVe 6B, padding_idx = 0) | Frozen (0 train.) |
| **Encoder Stack** | `nn.TransformerEncoder` | 4 × `TransformerEncoderLayer` | 869,192 |
| ↪ Multi-Head Attention | `nn.MultiheadAttention` | $d_{\text{model}} = 50$, $n_{\text{head}} = 5$ ($d_k = 10$ per head) | |
| ↪ Linear 1 | `nn.Linear` | in = 50 → out = 2048 (dim_feedforward) | |
| ↪ Dropout 1 & 2 | `nn.Dropout` | $p = 0.1$ | |
| ↪ Linear 2 | `nn.Linear` | in = 2048 → out = 50 | |
| ↪ LayerNorm 1 & 2 | `nn.LayerNorm` | Shape $(50,)$, $\epsilon = 10^{-5}$, elementwise affine | |
| **Sentence Pooling** | Temporal Mean Reduction | $\mathbf{u} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{h}_t \in \mathbb{R}^{50}$ | Param-free |
| **Classification Head** | `nn.Linear` | in = 150 ($3 \times d_{\text{model}}$) → out = 3 | 453 |

Table 1: Detailed structural topology of our Siamese SBERT neural network.

## 2.4 Program Flow and Training Pipeline

The program execution is divided into the following sequential stages:

- **Tokenization and Indexing**: Text is tokenized using regular expressions over Unicode word and punctuation boundaries (`r'\p{L}+|\p{N}+|\p{P}'`), normalized to lower case, and mapped to integer sequences via `token2idx`. Class targets are mapped via `label2idx` (contradiction = 0, entailment = 1, neutral = 2).
- **Forward Pass and Dual Purpose**: During training, `forward(u, v)` runs sentences $s_1$ and $s_2$ through the shared Siamese encoder. Mean pooling over sequence lengths yields vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{50}$. The combined vector $[\mathbf{u}, \mathbf{v}, |\mathbf{u}-\mathbf{v}|] \in \mathbb{R}^{150}$ is projected by the linear head to output 3-class logits. In inference mode, `encode_sbert()` detaches the classification head and returns pure 50-dimensional contextual sentence embeddings.
- **Optimization Routine**: Samples are processed individually ($B = 1$) to eliminate artificial batch padding noise. Training utilizes cross-entropy loss (`nn.CrossEntropyLoss`) and the Adam optimizer (lr $= 2 \times 10^{-5}$) over 5 epochs.

# 3 Results

## 3.1 Quantitative Results on SNLI

We evaluate the embedding quality by calculating the mean pairwise cosine similarity between premise and hypothesis across each logical relation in the dataset. Table 2 presents the empirical results.

| Method | Entailment | Neutral | Contradiction | Dynamic Range (Δ) |
|---|---|---|---|---|
| **GloVe Baseline (Mean-Pooling,** $50d$**)** | 0.9453 | 0.9385 | 0.9298 | 0.0155 |
| **Siamese SBERT (4-layer Transformer,** $50d$**)** | **0.6811** | **0.5483** | **0.3863** | **0.2948** |

Table 2: Mean cosine similarity between premise and hypothesis pairs on SNLI.

During SBERT training, the empirical 3-class cross-entropy loss decreased monotonically:

$$\text{Epoch 1: } 1.0416 \longrightarrow \text{Epoch 2: } 0.9616 \longrightarrow \text{Epoch 3: } 0.9211 \longrightarrow \text{Epoch 4: } 0.8939 \longrightarrow \text{Epoch 5: } 0.8645$$

confirming steady parameter convergence and gradient descent across all 4 encoder layers and the projection head.

### 3.2 Analysis of Baseline Vector Collapse vs. SBERT Disentanglement

The empirical results in Table 2 illuminate the fundamental difference between static word averages and contextual Siamese encoders:

- **GloVe Vector Collapse**: The baseline exhibits severe vector collapse: all three categories cluster within an ultra-narrow band between 0.9298 and 0.9453 (a separation gap $\Delta$ of merely 0.0155). In SNLI, pairs share frequent stopwords and high lexical overlap (e.g., "a person is walking..."). Because static averaging ignores word order, negation, and syntactic composition, the shared vocabulary vectors dominate the average direction, rendering naive GloVe entirely unable to tell whether two sentences agree or directly contradict each other.
- **SBERT Disentanglement**: By leveraging the 3-class auxiliary pretext loss, SBERT expands the dynamic separation range nearly twenty-fold ($\Delta = 0.2948$). Entailment pairs (semantically equivalent) achieve high similarity (0.6811), neutral pairs settle at intermediate values (0.5483), and contradiction pairs drop sharply to 0.3863. The auxiliary classification task successfully forces the shared Transformer to project incompatible assertions into distant geometric coordinates.

### 3.3 Sentence Embedding Inference and Generalization

To demonstrate that SBERT functions as a general sentence embedder rather than a mere pair classifier, we discard the 3-class head and apply `encode_sbert()` to three unseen standalone sentences:

- $S_1$: *"The weather is lovely today."*
- $S_2$: *"It's so sunny outside!"*
- $S_3$: *"He drove to the stadium."*

Computing the cosine similarity between their resulting 50-dimensional vectors yields the matrix:

$$\mathbf{S} = \begin{pmatrix} 1.0000 & \mathbf{0.7600} & 0.0818 \\ \mathbf{0.7600} & 1.0000 & 0.4713 \\ 0.0818 & 0.4713 & 1.0000 \end{pmatrix} \tag{5}$$

Despite $S_1$ and $S_2$ sharing virtually no vocabulary, SBERT maps them to strongly aligned representations with a cosine similarity of **0.7600**. In contrast, $S_3$ (an unrelated sports/driving statement) is mapped nearly orthogonally to $S_1$ (0.0818). This confirms that the pretext NLI training successfully instilled true contextual semantic understanding into the standalone embedder.

## 4 Conclusion

This assignment demonstrated that static word embedding averaging fails as a sentence representation method due to lexical dominance and vector collapse. By utilizing a 3-way NLI classification task as an auxiliary training scaffold, Sentence-BERT successfully sculpts a continuous geometric space where arbitrary sentences can be encoded into dense vectors. Once fine-tuned, the classifier is discarded, yielding an embedder whose cosine distances accurately measure semantic similarity across unseen text. Future enhancements include migrating to subword tokenization (WordPiece/BPE) and scaling to contrastive Multiple Negatives Ranking loss over large batch distributions.

## References


[1] N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992, 2019.

[2] S. R. Bowman, G. Angeli, C. Potts, and C. D. Manning. A large annotated corpus for learning natural language inference. In *Proceedings of EMNLP*, pages 632–642, 2015.

[3] J. Pennington, R. Socher, and C. D. Manning. GloVe: Global vectors for word representation. In *Proceedings of EMNLP*, pages 1532–1543, 2014.